

Saksham Checker

+44 7423127042 | sakshamchecker.com | linkedin.com/in/sakshamchecker | checkersaksham01@gmail.com

SUMMARY

Data Scientist with an MSc in Advanced Computing from King's College London and a strong background in Machine Learning research and deployment. Experienced in building LLM-driven analytical workflows, optimizing cloud-native ML pipelines on AWS, and developing Federated Learning solutions with a research focus on interpretable and trustworthy AI systems for decision-making in complex, real-world environments. Proven track record of publishing in high-impact journals and conferences.

EDUCATION

King's College London, UK

Sept 2024 – Sept 2025

MSc Advanced Computing — Grade: 73% (Distinction)

- **Modules:** Machine Learning, Deep Learning, Big Data Technologies, Agents & Multi-Agent Systems, Computer Vision, Distributed Systems, Data Mining.
- **Dissertation:** *Action Recognition and Quality Assessment for Figure Skating* – Developed a ViT-based pipeline for fine-grained action classification and automated scoring in sports.

Delhi Technological University (DTU), India

Aug 2020 – June 2024

B.Tech in Information Technology — Grade: 8.77/10.0 CGPA

- **Modules:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Artificial Intelligence, Software Engineering, Discrete Mathematics.
- **Final Project:** *Classification of Crystal Structure of Perovskite Oxide* – Developed hybrid machine learning models (**EDKSVM, LightGBM**) for **crystal structure classification**, achieving **98% accuracy** on large-scale datasets.

EXPERIENCE

Penta Group — London, UK

Oct 2024 – Present

Associate Data Scientist (Previously Graduate Data Science Developer)

- Designed and deployed **LLM-driven analytical systems** within **containerized environments**, enabling **scalable inference** for large-scale text data and supporting **decision-making in complex, real-world settings**.
- Developed and fine-tuned **Large Language Models (LLMs)** for tasks such as **sentiment analysis** and transcript understanding, with a focus on improving **model reliability, consistency, and interpretability**.
- Built a **real-time information analysis pipeline** processing **10,000+ articles daily**, applying **topic modelling (LDA, NMF)** and **semantic clustering** to identify evolving narratives and support **situational awareness under uncertainty**.
- Optimised **inference and data pipelines**, achieving a **40% reduction in compute cost**, while ensuring **scalable and reproducible deployment** of AI systems in production environments.

University of Cambridge — Cambridge, UK

Mar 2023 – Feb 2024

Visiting Student (Affective Intelligence and Robotics Lab)

- Engineered **FedLGR (Federated Latent Generative Replay)**, a framework that enables decentralized robot fleets to learn new social behaviors without "catastrophic forgetting" of prior tasks.
- Validated the architecture on the **MANNERS-DB** dataset, achieving a PCC score of 0.58 and demonstrating superior stability compared to standard Federated Averaging (FedAvg).
- Contributed to the design of **FedRoot**, an end-to-end Federated Learning solution for low-resource robotic hardware.

Delhi Technological University — New Delhi, India

Dec 2020 – June 2024

Researcher (Big Data and Web Analytics Lab)

- Conducted advanced ML research under the guidance of **Prof. Dr. Rahul Katarya**, focusing on cybersecurity and NLP.
- Developed **DIMC**, a hybrid deep learning model (DenseNet + InceptionV3) that transforms executable binaries into images for malware classification, reaching **99.14% accuracy** on the MallImg dataset.
- Led the curation of **HiEnWrite**, a novel bilingual (Hindi-English) dataset from 400 authors to study **Big Five personality traits** via graphology and end-to-end Convolutional Networks.
- Authored and published high-impact research on **Firefly-optimized** fake news detection and crystal structure classification for material science in **IEEE Access** and **ACM Transactions**.

PUBLICATIONS AND CONFERENCES

[1] **Fake News Detection System Using Feature-Based Optimized ML Classification**

IEEE Access, vol. 10, pp. 116074-116087, 2022. DOI: [10.1109/ACCESS.2022.3216892](https://doi.org/10.1109/ACCESS.2022.3216892)

- Developed a hybrid optimization framework using the **Firefly Algorithm** and **Principal Component Analysis (PCA)** for feature selection.
- Achieved **90–99% accuracy** across benchmarks (PolitiFact, GossipCop) using Multi-Class SVMs, outperforming several state-of-the-art baselines.

[2] **HiEnWrite: A Hindi-English Bilingual Dataset for Big Five Personality Detection**
ACM Transactions on Asian and Low-Resource Language Processing, 2024. DOI: [10.1145/3756010](#)

- Curated the first-of-its-kind bilingual handwritten dataset from **400 authors** to bridge the gap in low-resource language personality research.
- Implemented an **End-to-End CNN architecture** to extract graphological features, mapping handwriting strokes to the Big Five personality traits.

[3] **Exploring the Classification of Crystal Structure of Perovskite Oxide Using Machine Learning and Deep Learning**
IEEE International Conference on Computer, Communication and Control Systems (IC3SE), 2024. DOI: [10.1109/ELEXCOM58812.2023.10370067](#)

- Engineered a custom **Encoder-Decoder Kernel SVM (EDKSVM)** using **TensorFlow** to classify perovskite structures (cubic, tetragonal, etc.).
- Optimized a **LightGBM** model on an augmented dataset of 24,990 material points, achieving **98% classification accuracy** for materials discovery.

[4] **A Survey on Intrusion Detection System for IoT Networks Based on Artificial Intelligence**
IEEE ELEXCOM, Roorkee, India, 2023. DOI: [10.1109/ELEXCOM58812.2023.10370067](#)

- Conducted a comprehensive survey of **anomaly-based IDS**, analyzing the trade-offs between LSTM, GRU, and traditional Random Forest models in high-traffic networks.

[5] **Solar Panels Crack Detection using Overhead Images**
International Journal for Research in Applied Science and Engineering Technology (IJRASET), 2021. DOI: [10.22214/ijraset.2021.38532](#)

- Achieved **95.34% accuracy**, enabling automated fault detection and improving maintenance efficiency in solar energy systems.

[6] **Federated Continual Learning for Socially Appropriate Robot Behaviours**
Preprint. arXiv:2405.15773 [cs.RO]
LEAP HRI - 2024, USA, March 2024 [arXiv:2403.07586 \[cs.RO\]](#)

- Architected **FedLGR**, utilizing the **Flower.ai** framework to implement Latent Generative Replay for decentralized robotics.
- Demonstrated that generative replay significantly mitigates catastrophic forgetting in non-IID (non-independent and identically distributed) data streams across robot fleets.

PROJECTS

DIMC: Malware Classification

DTU — Guided by Prof. Rahul Katarya

- Developed a novel deep learning architecture for high-accuracy malware classification, leveraging a hybrid approach combining DenseNet and InceptionV3 models to analyze executable binary images.
- Achieved state-of-the-art classification performance in identifying 25 distinct malware families, demonstrating 98.20% accuracy on the MaleVis dataset and an exceptional 99.14% accuracy on the MalImg dataset.

Automated Figure Skating Assessment

King's College London — MSc Project

- Developed and implemented a novel end-to-end pipeline for automated action recognition and quality assessment in figure skating, processing video data to provide objective performance analysis.
- Engineered a Vision Transformer (ViT) based architecture to capture long-range temporal dependencies, accurately classifying 15 fine-grained actions (e.g., jumps, spins) and demonstrating expertise in temporal action localization.

ACHIEVEMENTS

- **Vice Chancellor's Award Scholarship:** Awarded GBP 10,000 for postgraduate excellence at King's College London.
- **Toycathon 2021 Finalist:** Ranked in Top 10 nationally by the Ministry of Education, India.

LEADERSHIP ROLES

President and ML Mentor, Round Table DTU

July 2022 – July 2023

- Led a technical society of over 300 members during the academic year of 2022-23
- The society completed projects and organized national-level events with a total of more than 800 participants throughout the year

SKILLS

- **Languages:** Python, SQL, C++, Bash.
- **AI/ML:** PyTorch, TensorFlow, Keras, Scikit-learn, LLMs (Fine-tuning/RAG), Computer Vision, NLP.
- **Cloud & DevOps:** AWS (SageMaker, Lambda, ECS, ECR), Docker, CI/CD, Git, MLOps.
- **Data Engineering:** Pandas, NumPy, LDA, NMF Topic Modelling.